

Practice effects distort translational validity estimates for a Neurocognitive Battery.

ABSTRACT

BACKGROUND: When a rating instrument is translated into a different language, estimates of translational validity are typically obtained by administering the test in the original and the translated versions to individuals familiar with both languages. We investigated whether and how practice effects affect the translational validity estimates.

METHODS: The Penn neurocognitive battery (Penn CNB) is a semi-automated instrument for estimating speed and accuracy of performance in several cognitive domains. To develop an Arabic version of the CNB, the English Penn CNB was translated into Arabic, then back translated and revised. The Arabic and the original English versions were administered in a randomized crossover design to bilingual Arab participants (N=22).

RESULTS: Performance varied by cognitive domain, but it generally improved at the second session regardless of the language of the initial test. When performance on the English and Arabic versions was compared, significant positive correlations were detected for accuracy in 8/13 cognitive domains and for speed in 4/13 domains ($r=0.02-0.97$). The practice effect for each variable was estimated using linear models. When the practice estimates were incorporated, the translational validity estimates improved substantially (accuracy, $r=0.50-0.96$, speed, $r=0.63-0.92$, all correlations, $p=0.05$ or better).

Conclusion: Translational validity estimates for cognitive measures can be distorted by practice effects, if there is inter-individual variability.

INTRODUCTION

Computer based testing for cognitive functions, has come to the fore because it enables flexibility in test choice, provides accurate estimates for time to completion, and eases scoring, data storage and access (**Cernich et al., 2007**). To enable accurate cognitive testing for individuals with severe psychiatric disorders, Gur et al (2001) described the Penn neurocognitive battery (Penn CNB) that enables accuracy as well as speed for a range of cognitive domains (**Gur et al., 2001b**). The Penn CNB reliably sex differences & age gradients among research participants (**Gur et al., 2001b**). It has acceptable construct validity (**Gur et al., 2010**) and the well documented cognitive impairments in schizophrenia are also detectable (**Gur et al., 2001a**). The Penn CNB has been successfully implemented in large-scale genetic studies that have enabled estimates of heritability as well as gene mapping studies of cognitive domains (**Greenwood et al., 2007; Gur et al., 2007a; Gur et al., 2007b**). It is now publicly available and can be administered through the internet (<https://pennncnp.med.upenn.edu>). Thus, large-scale normative and disease-specific data on thousands of individuals are available. The Penn CNB have been translated into over twenty languages (**Deshpande et al., 1998; Gur et al., 2010**). We describe an Arabic version of the CNB. We report on distortions due to practice effects when estimating translational validity (**Inzelberg et al., 2007**).

METHODS

Study design: The English version of the Penn CNB was translated into Arabic by bi-lingual faculty level psychiatrists at Mansoura University Hospital (MUH), Mansoura, Egypt. Subsequently, the Arabic version was back translated by a different team of psychiatrists. The original and the back translated versions were compared for discrepancies that were corrected. Next, the revised Arabic and the original English versions of the Penn CNB were administered to Arab individuals proficient in English and Arabic. Half of the participants completed the Arabic versions of the CNB first, followed by the English version, while the rest completed the English version first. The study was approved by the Institutional Review Boards (IRB) at Mansoura University and the University of Pittsburgh. Verbal consent was obtained from all participants, per Mansoura University and the University of Pittsburgh IRB regulations.

Session of the Penn CNB: The battery was administered in a quiet space convenient for each participant. The battery was re-administered at the same place and at approximately the same time of day as the initial test. The following tests were included.

1. Motor Praxis Test (MPRAXIS).
2. Penn Continuous Performance Test – Number (PCPT-n)
3. Penn Translational Memory Test (CPF).
4. Penn Translational Memory Test Delayed Memory (CPFd).
5. Penn Conditional Exclusion Task (PCET).
6. Short Computerized Finger-Tapping Task (sCTAP).
7. Short Visual Object Learning Test (sVOLT).
8. Short Visual Object Learning Test Delayed Memory (sVOLTd).
9. Measured Emotion Differentiation Test (MEDF).
10. Penn Matrix Reasoning Test (PMAT).
11. Short Penn Line Orientation Test (sPLOT).
12. Age Differentiation Test (ADT).
13. Penn Emotion Recognition Task (ER40).
14. Short Fractal N-Back (SFNB2).

MPRAXIS was used primarily to familiarize participants with the computer intertranslational, so it was not included for data analysis.

Statistical analysis: The Statistical Package for Social Sciences (SPSS, version 19.0 for Windows) and R version 2.15.1 was used for all analyses.

RESULTS

Demographic details. There were 22 participants (16 women and 6 men). All the participants were high school graduates (years of education, mean $\pm$ standard deviation, 17.59 ± 2.08 years, range 12-20). The mean age of participants was 30.18 ± 9.14 years (range 18-48). The mean interval between the first and second session was 14.91 ± 1.5 days (range 13 to 19 days).

Completion rates. Each participant completed the battery in less than 1.5 hours without breaks, except for one participant.

Mean difference. Paired t-tests showed no statistically significant differences between the English and Arabic versions for any of the cognitive tests either with regard to accuracy or completion times, regardless of the order of administration (Supplementary Table 1s, 2s, 3s).

Correlations between English and Arabic versions of the Penn CNB.

(i) Translational validity unadjusted for practice effects.

Accuracy: A significant positive correlation was detected for 8 out of the 13 comparisons between the English and Arabic versions ($p = 0.05$, Table 1). Overall, the range of correlation values was extensive ($r = 0.02-0.97$).

Speed: Only 4 were statistically significant were observed when speed variables were compared for each cognitive domain ($p = 0.05$ or better, Table 1). The range of correlations was as broad as that observed for accuracy ($r = 0.03 - 0.8$). There is no time component for the Short Computerized Finger-Tapping Task, as participants are asked tap as many times as possible in 60 seconds.

Table 1. Correlations for performance in the Arabic and English versions of the Penn CNB.

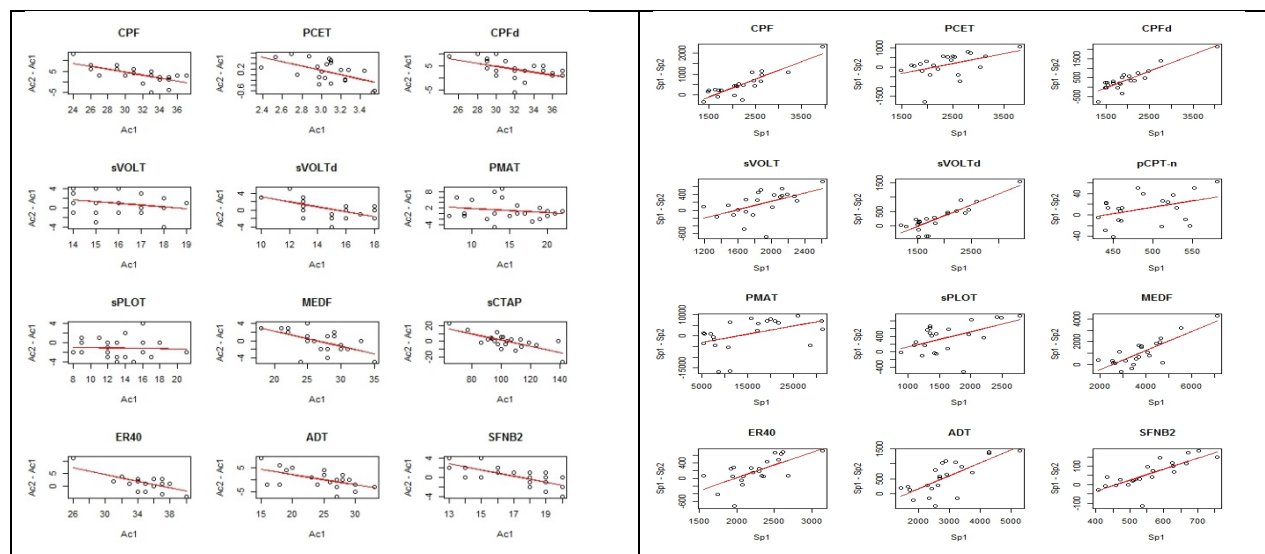
Test	Abbreviation	Accuracy		Speed	
		Correlation	p	Correlation	p
Penn Translational Memory Test Delayed	CPFd	0.02	0.936	0.03	0.896
Penn Translational	CPF	0.07	0.769	-0.03	0.885
Penn Conditional Exclusion Task	PCET	0.14	0.539	0.41	0.058
Short Visual Object Learning Test	sVOLT	0.33	0.131	0.42	0.053
Penn Emotion Recognition Task	ER40	0.37	0.090	0.32	0.144
Short Visual Object Learning Test Delayed	sVOLTd	0.44	0.041	0.27	0.232
Short Fractal N-Back	SFNB2	0.50	0.017	0.38	0.085
Age Differentiation Test	ADT	0.67	0.001	0.60	0.003
Penn Matrix Reasoning Test	PMAT	0.70	<0.001	0.68	0.001
Measured Emotion Differentiation Test	MEDF	0.79	<0.001	-0.08	0.735
Short Penn Line Orientation Test	sPLOT	0.80	<0.001	0.49	0.021
Short Computerized Finger-Tapping Task	sCTAP	0.82	<0.001	-----	-----

Penn Continuous Performance Test –	pCPT-n	0.97	<0.001	0.8	0.000
------------------------------------	--------	------	--------	-----	-------

(ii) Translation validity adjusted for practice effects.

We observed that most of the cognitive tests, individuals performed better at the second session, compared with the first session. The change in accuracy estimates between the second and the first session ($Ac2-Ac1$) appeared to be a linear function of performance at the first session ($Ac1$, Figure 1, left panel). In other words, individuals who performed better at the first session showed smaller improvement compared with individuals who performed worse. Speed typically improved at the second session (ie, $Sp2$, the second observation) was numerically smaller than $Sp1$, the first observation. For clarity, the difference in speed estimates between the first and the second session ($Sp1-Sp2$) were plotted against the speed estimates for the first session (Figure 1, right panel). The trends for the speed measures also suggested the presence of practice effects. However, the relationship of $Ac2 - Ac1$ versus $Ac1$ for the Penn Continuous Performance Test was the converse of the trends for the other cognitive tests (pCPT-n, Figure 2, left panel). When an outlier was removed from the analyses (Figure 2, bottom left of left panel), the slope $Ac2-Ac1$ appeared to be consistent with the other cognitive tests (Figure 2, right panel).

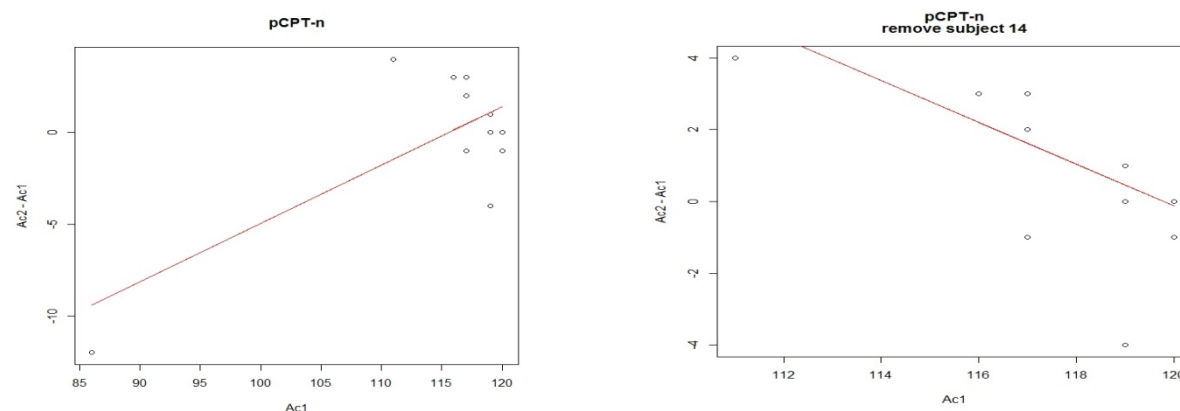
Figure 1. Change in accuracy and speed between the first and second session of the Penn CNB as a function of performance at the first session.



The slope of change in accuracy (left panel) and speed are shown (right panel).

$Ac1$, $Ac2$: accuracy of the test in the first and second test sessions, respectively; $Sp1$, $Sp2$: speed of the test in the first and second test sessions respectively. For explanation of abbreviations of cognitive tests, please see second column of Table.

Figure 2. Accuracy estimates for the Penn Continuous Performance test.



Accuracy estimates for the first and second sessions of the Penn continuous Performance test are shown ($Ac1$, $Ac2$, respectively) for the entire sample (left panel). In the right panel they are shown after removal of an outlier in the left lower panel.

pcpt: Penn continuous Performance test.

Therefore the practice effect was modeled as follows. For subject i , we fit the linear model on the other $n-1$ subjects, and then predicted the learning effect for subject i . Then we computed the correlations between Arabic and English versions, after removing the practice effect predicted from the linear models. Table 2 shows correlations of Arabic and English after adjusting for the predicted learning effects. The correlations improved substantially for several variables and the range of values appeared smaller in comparison with the unadjusted correlations. All the correlations were statistically significant ($p = 0.05$ or better).

Table 2: Correlations after adjusting for practice effects.

Test	Abbreviations	Accuracy		Speed	
		Correlation	p	Correlation	p
Penn Translational Memory Test Delayed Memory	CPFd	0.61	0.003	0.90	<0.001
Penn Translational Memory	CPF	0.71	<0.001	0.89	<0.001
Penn Conditional Exclusion Task	PCET	0.58	0.005	0.63	0.002
Short Visual Object Learning Test	sVOLT	0.50	0.017	0.74	<0.001
Penn Emotion Recognition Task	ER40	0.75	<0.001	0.79	<0.001
Short Visual Object Learning Test Delayed Memory	sVOLTd	0.74	<0.001	0.90	<0.001
Short Fractal N-Back	SFNB2	0.84	<0.001	0.90	<0.001
Age Differentiation Test	ADT	0.82	<0.001	0.92	<0.001
Penn Matrix Reasoning Test	PMAT	0.73	<0.001	0.76	<0.001
Measured Emotion Differentiation Test	MEDF	0.88	<0.001	0.84	<0.001
Short Penn Line Orientation Test	sPLOT	0.83	<0.001	0.84	<0.001
Short Computerized Finger-Tapping Task	sCTAP	0.91	<0.001	-----	-----
Penn Continuous Performance Test – Number	pCPT-n	0.96	<0.001	0.84	<0.001

DISCUSSION:

While developing an Arabic version of the Penn CNB, we observed unsatisfactory correlations between cognitive variables estimated during two sessions. The changes in the cognitive variables varied between individuals and across cognitive domains, but were related inversely to the estimates from the first session, consistent with practice/learning effects that were modeled and estimated. Upon adjustment for the practice estimates, the correlations between the first and the second sessions improved substantially. Our results are consistent with another study in which test-specific practice effects on performance in cognitive domains were estimated through repetitive testing using the English version of a cognitive battery over one

year. Improvement from baseline to second testing approximately 2-3 weeks later accounted for the largest proportion of change in all cognitive domains, most prominently for executive functions and learning/memory (**Bartels et al., 2010**). After three months and upon reduced testing frequency over one year, the acquired cognitive level stabilized (**Bartels et al., 2010**). A reliable prediction of expected practice effects under healthy good-performance conditions is thus deemed essential for accurate effect size estimation and for selection of a suitable test set (**Bartels et al., 2010**). To our knowledge the impact of practice estimates on translational validity estimates have not been reported.

To avoid overfitting, we adjusted the correlations for the practice effect by utilizing the predicted value, not the fitted value, but other relevant variables could impact the magnitude of the practice effect estimates. Evaluation of a larger sample is desirable in order to obtain more accurate estimates of the variability. The Practice effect may also be overestimated in our sample due to the short interval between the first and second session, the relatively young sample (range: 18-48 years) and relatively high educational achievements of the participants (average: 17 years). Practice effects was not detected when the Penn CNB was administered after a 1-2 year interval (**Gur et al 2001**) (**Roalf et al, 2014**), but practice-induced score gains are well documented for other cognitive tests when they were re-administered following a shorter time lag (**Hausknecht et al., 2007**). Since the order of language version was randomized among subjects, we do not expect the language order to be associated with the initial performance.

In conclusion, we report on an Arabic version of the Penn CNB that can be used for a range of research requiring assessment of cognitive function. When testing the validity of other translated versions of cognitive tests, we recommend adjustment for practice effects.

ACKNOWLEDGEMENTS: This study was funded in part by support from NIMH, the Fogarty International Center (MH 63480, MH093246, D43TW008302, D43 TW009114) and the Stanley Medical Research Institute (07R-1712).

REFERENCES:

Baker, E. L., Letz, R. E., Fidler, A. T., Shalat, S., Plantamura, D. & Lyndon, M. (1985) A computer-based neurobehavioral evaluation system for occupational and environmental epidemiology: methodology and validation studies. *Neurobehav Toxicol Teratol*, 7 (4), 369-377.

- Bartels C., Wegrzyn M., Wiedl A., Ackermann V., Ehrenreich H.: Practice effects in healthy adults: A longitudinal study on frequent repetitive cognitive testing. *BMC Neuroscience* 2010, 11:118.
- Bearden CE, Hoffman KM, Cannon TD. The neuropsychology and neuroanatomy of bipolar affective disorder: A critical review. *Bipolar Disord.* 2001;3:106–50.
- Cernich A.N., Brennana D.M., Barker L.M. and Bleiberg J., Sources of error in computerized neuropsychological assessment, *Arch Clin Neuropsychol* 22(Suppl 1) (2007), 39–48.
- Deshpande, S. N., Mathur, M. N., Das, S. K., Bhatia, T., Sharma, S. & Nimgaonkar, V. L. (1998) A Hindi version of the Diagnostic Interview for Genetic Studies. *Schizophrenia Bulletin*, 24 (3), 489-493.
- Greenwood TA, Braff DL, Light GA, Cadenhead KS, Calkins ME, Dobie DJ, et al. Initial heritability analyses of endophenotypic measures for schizophrenia: the consortium on the genetics of schizophrenia. *Arch Gen Psychiatry* 2007;64:1242–50.
- Gur RE, Calkins ME, Gur RC, Horan WP, Nuechterlein KH, Seidman LJ, et al. The Consortium on the Genetics of Schizophrenia: neurocognitive endophenotypes. *Schizophr Bull* 2007a;33:49–68.
- Gur RE, Nimgaonkar VL, Almasi L, Calkins ME, Ragland JD, Pogue-Geile MF, et al. Neurocognitive endophenotypes in a multiplex multigenerational family study of schizophrenia. *Am J Psychiatry* 2007b;164:813–9.
- Gur, R. C., Ragland, J. D., Moberg, P. J., Bilker, W. B., Kohler, C., Siegel, S. J. & Gur, R. E. (2001a) Computerized neurocognitive scanning: II. The profile of schizophrenia. *Neuropsychopharmacology*, 25 (5), 777-788.
- Gur, R. C., Ragland, J. D., Moberg, P. J., Turner, T. H., Bilker, W. B., Kohler, C., Siegel, S. J. & Gur, R. E. (2001b) Computerized neurocognitive scanning: I. Methodology and validation in healthy people. *Neuropsychopharmacology*, 25 (5), 766-776.
- Gur, R. C., Richard, J., Hughett, P., Calkins, M. E., Macy, L., Bilker, W. B., Brensinger, C. & Gur, R. E. (2010) A cognitive neuroscience-based computerized battery for efficient measurement of individual differences: standardization and initial construct validation. *J Neurosci Methods*, 187 (2), 254-262.
- Gur, R. C., Saykin, A. J., Benton, A., Kaplan, E., Levin, H., Kester, D. B. & Gur, R. E. (1990) "Behavioral Imaging"-III. Interexpert Agreement and Reliability of Weightings. *Neuropsychiatry, Neuropsychology, & Behavioral Neurology Summer*, 3 (2), 113-124.
- Hausknecht JP, Halpert JA, Di Paolo NT, Gerrard MOM: Retesting in selection: A meta-analysis of coaching and practice effects for tests of cognitive ability. *J Appl Psychol* 2007, 92:373-385.

- Inzelberg, R., Schechtman, E., Abufel, A., Masarwa, M., Mazarib, A., Strugatsky, R., Farrer, L. A., Green, R. C. & Friedland, R. P. (2007) Education effects on cognitive function in a healthy aged Arab population. *Int Psychogeriatr*, **19** (3), 593-603.
- Martinez-Aran A, Penades R, Vieta E, et al. Executive function in patients with remitted bipolar disorder and schizophrenia and its relationship with functional outcome. *PsychotherPsychosom*. 2002;71:39–46.
- Roalf DR, Gur RC, Almasy L, Richard J, Gallagher RS, Prasad K, Wood J, Pogue-Geile MF, Nimgaonkar VL, and Gur RE. Neurocognitive Performance Stability in a Multiplex Multigenerational Study of Schizophrenia. *Schizophr Bull* (2013) 39 (5): 1008-1017.
- Sidney J. Segalowitz, Patrick Mahaney, Diane L. Santesso, Leslie MacGregora, Jane Dywana and Barry Willer. computerized neuropsychological battery used to assess recovery from concussion. *NeuroRehabilitation* 22 (2007) 243–251.
- Woods SP, Lovejoy DW, Ball JD. Neuropsychological characteristics of adults with ADHD: A comprehensive review of initial studies. *ClinNeuropsychol*. 2002;16:12–34.